A coupled panel method and hydraulic network for porous airfoil aerdynamics[⋆,⋆⋆]

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ABSTRACT

Passive, pressure-driven flow between connected surface pores can modify aerodynamic forces without external actuation. This study develops a reduced-order aerodynamic–hydraulic framework for a NACA 0018 airfoil by coupling an inviscid source–vortex panel method with rectangular-channel flow models. Three configurations are investigated: horizontal channels connecting chordwise locations on the same surface, vertical channels connecting the lower and upper surfaces, and a combined arrangement of hydraulically independent channels. Aerodynamic coefficients are obtained by integrating the predicted surface pressure. At positive angles of attack, horizontal channels increase lift through pressure redistribution, with larger channel heights producing greater increases in lift. Vertical channels reduce the pressure difference between the two surfaces and decrease lift relative to the impermeable baseline. In the combined configuration, the channels interact through the external pressure field, yielding a response closer to that of the solid airfoil. Pressure and velocity difference fields show that the strongest flow modifications occur near the pore openings. These results identify channel orientation and pore connections as key factors governing the aerodynamic response of porous airfoils. The framework supports preliminary design screening and provides inviscid predictions for subsequent assessment using viscous simulations and experiments.

1. Introduction

Porous wings have attracted attention due to their favourable acoustic properties [1]. Beyond acoustic applications, airfoils with anisotropic permeability have been considered for aerodynamic benefits, including lift and efficiency enhancement [2, 3]. Due to the multiscale nature of these flows, the design of porous wings remains challenging. This study addresses this challenge through a computationally inexpensive, extended panel-method formulation.

Previous porous-airfoil studies have reported reduced lift and either increased or decreased drag compared with solid airfoils [4, 5]. Experimental and numerical evidence shows that aerodynamic performance depends on the location of the porous region and the permitted internal flow directions [2, 1, 3]. These findings motivate the anisotropic and heterogeneous pore configurations examined in the present study. Here, anisotropy refers to the permitted directions of internal flow imposed by the channels. Heterogeneity refers to the spatial distribution of permeable pores and impermeable regions over the airfoil surface.

Thin airfoil theory provides efficient predictions of steady and unsteady porous airfoil behaviour [6, 7]. These models describe how the pressure difference between the lower and upper airfoil surfaces drives flow through the porous regions. Predicted surface pressures may differ from measurements when the porous region covers only a small part of the chord or offers little resistance to flow [8]. These models neglect airfoil thickness and do not explicitly describe internal channels connecting selected surface pores. Representing such channels requires the pressure at both pores and a relation linking channel dimensions and pressure difference to flow rate.

Passive internal channels transfer fluid between surface locations through pressure differences. Selected slot geometries can increase the peak lift-to-drag ratio and improve aerodynamic performance at high angles of attack [9, 10]. This internal flow can also alter separated vortex structures [11]. However, this study focuses on specific geometries. A general reduced-order model is needed to compare multiple independent channels connecting selected pairs of surface pores.

Porous region optimisation has been investigated for static-stall control [12]. For airfoils with individual channels, the design must also specify which pores are connected and the dimensions and orientation of each channel. Channel length and height govern hydraulic resistance, while pore locations affect the driving pressure difference. These variables must therefore be assessed together. Evaluating their combinations through viscous simulations or experiments is costly, motivating an efficient model for preliminary design.

Panel methods efficiently calculates surface velocities and pressures while accounting for airfoil thickness. Only the airfoil surface is discretised, avoiding a mesh of the surrounding fluid [13, 14, 15]. In a two dimensional source–vortex formulation, the source strengths and circulation are

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determined by imposing the surface normal velocity condition and the trailing-edge Kutta condition. A prescribed normal velocity represents flow through each pore, while pressure differences between connected pores drive the internal channel flow. This links the aerodynamic and hydraulic models within an efficient computational framework. The external solver remains inviscid to examine the aero–hydraulic interaction at low computational cost. XFOIL provides a reference for the solid-airfoil baseline [16], while the coupled porous model requires separate validation.

A reduced-order framework is therefore needed to represent individual internal channels and their interaction with the external flow. The present study couples a two-dimensional source–vortex panel method to a hydraulic-resistance model. Each channel connects a selected pair of surface pores, with its flow determined by the pressure difference between them. This approach allows different pore connections to be assessed within a common aerodynamic model.

Three channel layouts are examined for a NACA 0018 airfoil. The horizontal layout contains four channels connecting pores at different chordwise locations on the same surface. The vertical layout contains four channels connecting the lower and upper surfaces. The combined layout includes all eight channels as hydraulically independent connections that interact through the external pressure field. Channel heights of 8, 9, and 10 mm are assessed over angles of attack from $-5°$ to $15°$ at a chord Reynolds number of 5×10^5.

The contribution is a coupled aero–hydraulic model of channels connecting selected pore pairs. Each channel is hydraulically independent. This allows different channel connections to be compared using the same aerodynamic formulation. The framework relates channel geometry and connectivity to internal flow, surface pressure, lift, and pitching moment. It is intended for rapid preliminary design screening. Since boundary layers, skin friction, separation, and stall are not resolved, the predictions represent inviscid trends requiring subsequent viscous simulation and experimental validation.

2. Coupled aero-hydraulic methodology

2.1. Model overview

The framework couples the external flow around the airfoil to internal flow through channels connecting selected surface pores. Each channel is hydraulically independent. The channels interact through the external pressure field, while their flows impose suction and blowing at the pores.

The solution procedure is summarised in Fig. 1. For each geometry, the panel matrix is factorised once. The effect of unit flow through each channel is then precomputed. These coefficients allow surface velocities and pore pressures to be evaluated for trial channel flows without repeating the panel solve. This step is exact within the discretised aerodynamic model.

Initial channel flows are estimated from the impermeable-airfoil pressure field. At each iteration, the hydraulic model predicts the flows supported by the available pressure heads. The differences between the trial and predicted flows define the residuals. A damped Newton method updates all channel flows simultaneously [17]. The Jacobian is evaluated analytically.

The final candidate is checked independently using a full panel solve and the original hydraulic model. Acceptance requires a sufficiently small flow mismatch, a consistent energy balance and an admissible flow direction. The accepted surface pressures are then integrated to calculate the lift, pressure-drag and pitching-moment coefficients.

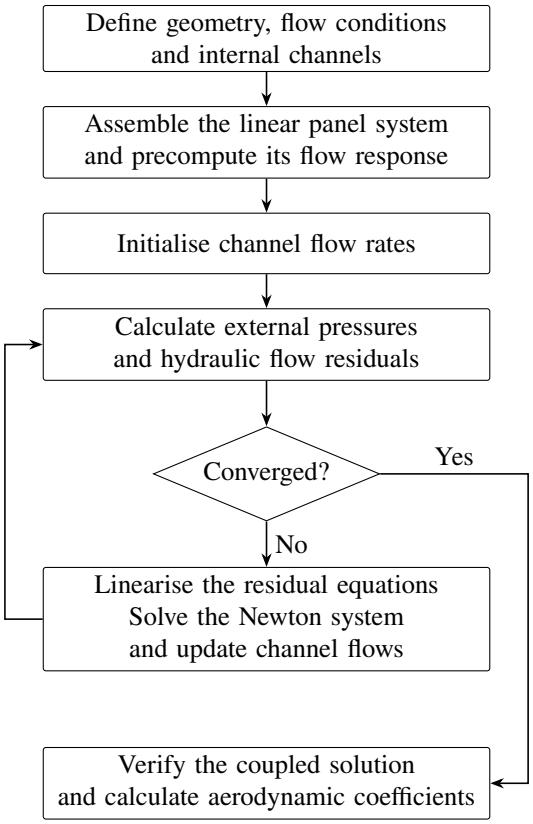


Figure 1: Coupled aero-hydraulic solution procedure. The linear panel equations are reduced exactly, while the nonlinear channel-flow residuals are linearised at each Newton iteration. Candidate solutions are independently verified before acceptance.

2.2. External source–vortex panel formulation

The external flow is assumed steady, incompressible, inviscid and irrotational in the fluid domain. The velocity is expressed using a potential Φ, such that $\mathbf{V} = \nabla\Phi$. Incompressibility requires $\nabla \cdot \mathbf{V} = 0$. The potential therefore satisfies Laplace's equation [18]:

$$\nabla^2\Phi = \frac{\partial^2\Phi}{\partial x^2} + \frac{\partial^2\Phi}{\partial y^2} = 0. \tag{1}$$

For a freestream speed V_∞ and angle of attack α, the freestream velocity and potential are

$$\mathbf{V}_\infty = V_\infty \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix}, \qquad \Phi_\infty = V_\infty(x\cos\alpha + y\sin\alpha). \tag{2}$$

The far-field condition is $\mathbf{V} \rightarrow \mathbf{V}_\infty$. The linearity of Laplace's equation allows the external flow to be represented by superposing the freestream with source and vortex distributions.

The airfoil boundary is discretised into N straight panels, ordered clockwise. Boundary conditions are imposed at the panel midpoints. Each panel carries a constant source strength λ_j. A uniform vortex-sheet strength γ is shared by all panels and permits nonzero circulation. This follows the source–vortex panel formulation [13, 14].

The total external velocity at position $\mathbf{r}$ is

$$\mathbf{V}(\mathbf{r}) = \nabla\Phi = \mathbf{V}_\infty + \sum_{j=1}^{N} \left[\lambda_j \mathbf{v}_j^{(s)}(\mathbf{r}) + \gamma \mathbf{v}_j^{(v)}(\mathbf{r})\right]. \quad (3)$$

Here, $\mathbf{v}_j^{(s)}$ and $\mathbf{v}_j^{(v)}$ are the velocities induced by unit-strength source and vortex distributions on panel j. Their signs follow the convention used below.

At collocation point i, the normal boundary condition is $\mathbf{V}_i \cdot \mathbf{n}_i = V_{n,i}$, where $\mathbf{n}_i$ is the outward unit normal. The prescribed velocity $V_{n,i}$ is zero on impermeable panels. At a channel opening, it represents the panel-average normal velocity caused by the internal flow. Positive values indicate blowing and negative values indicate suction. The normal-velocity equation is

$$\frac{\lambda_i}{2} + \frac{1}{2\pi} \sum_{\substack{j=1 \\ j\neq i}}^{N} I_{ij}\lambda_j - \frac{\gamma}{2\pi} \sum_{j=1}^{N} K_{ij} = V_{n,i} - V_\infty \cos\beta_i. \quad (4)$$

Here, β_i is the angle measured from the freestream direction to the outward panel normal. The tangential velocity, measured along the clockwise panel direction, is

$$V_{t,i} = V_\infty \sin\beta_i + \frac{1}{2\pi} \sum_{j=1}^{N} J_{ij}\lambda_j + \frac{\gamma}{2} - \frac{\gamma}{2\pi} \sum_{j=1}^{N} L_{ij}. \quad (5)$$

The coefficients I_{ij} and J_{ij} describe the normal and tangential source influences. The corresponding vortex coefficients are K_{ij} and L_{ij}. They are evaluated by analytical integration over each straight panel. Their diagonal entries are set to zero. The nonzero self-influence terms are included explicitly as $\lambda_i/2$ and $\gamma/2$.

The N normal-velocity equations are supplemented by a trailing-edge Kutta condition. For the adopted boundary ordering,

$$V_{t,1} + V_{t,N} = 0, \quad (6)$$

where panels 1 and N meet at the trailing edge. Together, these equations determine the source and vortex strengths.

For M channels, the flow vector is $\mathbf{q} = [q'_1, \ldots, q'_M]^T$, where q'_k is the flow rate per unit span through channel k. The prescribed normal velocities satisfy

$$\mathbf{V}_n = \mathbf{M}\mathbf{q}, \quad (7)$$

where $\mathbf{M}$ is the conservative transfer operator defined in Section **??**.

The aerodynamic unknowns are collected in $\mathbf{x} = [\lambda_1, \ldots, \lambda_N, \gamma]^T$. Multiplying the normal-velocity and Kutta equations by 2π gives

$$\mathbf{A}\mathbf{x} = \mathbf{b}_\infty + \mathbf{B}\mathbf{q}, \qquad \mathbf{B} = \begin{bmatrix} 2\pi\mathbf{M} \\ \mathbf{0}_{1\times M} \end{bmatrix}. \quad (8)$$

The matrix $\mathbf{A}$ contains the panel influence coefficients and the Kutta equation. The vector $\mathbf{b}_\infty$ contains the freestream contributions. The zero final row of $\mathbf{B}$ indicates that channel flow does not directly enter the right-hand side of the Kutta equation.

For prescribed channel flows, the aerodynamic system is linear. It is solved using an LU factorisation, which is reused for a fixed panel geometry. The resulting source and vortex strengths give the tangential velocities through Eq. (5).

Bernoulli's equation gives the pressure coefficient from the total local speed [18]:

$$C_{p,i} = 1 - \frac{V_{t,i}^2 + V_{n,i}^2}{V_\infty^2}. \quad (9)$$

Both velocity components are retained because the normal velocity can be nonzero at a pore opening. The panel pressure follows from $p_i = p_\infty + \frac{1}{2}\rho V_\infty^2 C_{p,i}$.

2.3. Internal hydraulic model

Each straight channel has height H_k and centreline length L_k, measured between its surface endpoints. Sidewall effects are neglected. The internal flow is represented between parallel walls using a two-dimensional, per-unit-span formulation. The signed flow rate q'_k is positive from opening 1 to opening 2.

The mean velocity, hydraulic diameter and channel Reynolds number are

$$U_k = \frac{q'_k}{H_k}, \quad (10)$$

$$D_{h,k} = 2H_k, \quad (11)$$

$$Re_{h,k} = \frac{\rho|U_k|D_{h,k}}{\mu}. \quad (12)$$

Here, ρ and μ are the fluid density and dynamic viscosity.

For steady, fully developed laminar flow, integration of the axial momentum equation with no-slip conditions at both walls gives the parallel-wall Hagen–Poiseuille relation [19]:

$$\Delta p_{f,k} = R'_k|q'_k| = \frac{12\mu L_k}{H_k^3}|q'_k|. \quad (13)$$

The quantity $\Delta p_{f,k} \geq 0$ is the magnitude of the distributed frictional pressure drop. The hydraulic resistance per unit span is $R'_k = 12\mu L_k/H_k^3$. For fixed geometry and fluid properties, the laminar frictional loss is therefore linear in the flow-rate magnitude.

The frictional pressure drop is expressed more generally using the Darcy–Weisbach relation [20]:

$$\Delta p_{f,k} = f_{D,k}\frac{L_k}{D_{h,k}}\frac{\rho U_k^2}{2}, \quad (14)$$

where $f_{D,k}$ is the Darcy friction factor. For parallel-wall laminar flow, $f_{D,k} = 96/Re_{h,k}$. Substitution into Eq. (14) recovers Eq. (13).

A two-regime closure is adopted. The laminar relation is used for $Re_{h,k} \le 2300$. Above this threshold, the smooth-pipe Blasius correlation is applied using the hydraulic diameter [20]:

$$f_{D,k} = \begin{cases} \dfrac{96}{Re_{h,k}}, & 0 < Re_{h,k} \le 2300, \\ 0.3164 Re_{h,k}^{-1/4}, & Re_{h,k} > 2300. \end{cases} \tag{15}$$

Applying the Blasius correlation to the parallel-wall geometry is an empirical approximation. The value 2300 is a prescribed switching threshold. No transition correlation or blending function is used. The resulting frictional pressure–flow relation is discontinuous at the threshold.

At zero flow, $\Delta p_{f,k}$ is set directly to zero. This avoids evaluating the singular friction-factor expression. During hydraulic inversion, the Reynolds number is recalculated from the candidate flow. A candidate is accepted only when it satisfies the channel energy equation and the Reynolds-number condition of its selected regime. If neither branch provides an admissible solution, the inversion is rejected.

2.4. Coupled formulation and nonlinear solution

The coupled unknowns are the signed channel flows $\mathbf{q} = (q'_1, \dots, q'_M)^\mathsf{T}$. For each channel, the opening with the higher average static pressure is selected as the inlet. Its normal kinetic head contributes to the driving head. Tangential kinetic energy is assumed to dissipate during entry.

For a nonzero opening-pressure difference and a uniform outlet jet ($\alpha_j = 1$), the channel energy balance is

$$\left|\Delta \overline{p}_k\right| + \frac{\rho}{2}\left\langle V_{n,\mathrm{in}}^2 \right\rangle = \Delta p_{f,k}(Q_k) + \frac{\rho}{2}\left(\frac{Q_k}{H_k}\right)^2, \tag{16}$$

where $\Delta \overline{p}_k = \overline{p}_{k,1} - \overline{p}_{k,2}$ and $Q_k = |q'_k|$. The brackets denote the opening average defined in Section **??**. The frictional loss $\Delta p_{f,k}$ follows Section 2.3. The outlet kinetic head is retained separately from this loss.

For given opening pressures and inlet normal kinetic head, hydraulic inversion determines the predicted flow magnitude. Its direction follows the sign of $\Delta \overline{p}_k$. At equal opening pressures, the model prescribes zero flow. Entrance turning and downstream jet mixing are not resolved explicitly.

For fixed geometry, the linear aerodynamic system in Eq. (8) is reduced before solving the channel-flow equations. A single LU factorisation of $\mathbf{A}$ is used to obtain

$$\mathbf{A}\mathbf{x}_0 = \mathbf{b}_\infty, \qquad \mathbf{A}\mathbf{X}_q = \mathbf{B}. \tag{17}$$

Here, $\mathbf{x}_0$ is the impermeable-airfoil solution. Each column of $\mathbf{X}_q$ gives the change in source and vortex strengths caused by unit flow through one channel. For any trial flow vector,

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{X}_q \mathbf{q}. \tag{18}$$

The surface velocities follow as

$$\mathbf{V}_t = \mathbf{V}_{t,0} + \mathbf{T}_q \mathbf{q}, \tag{19}$$

$$\mathbf{V}_n = \mathbf{M}\mathbf{q}, \tag{20}$$

where $\mathbf{V}_{t,0} = \mathbf{v}_{\infty,t} + \mathbf{C}\mathbf{x}_0$ and $\mathbf{T}_q = \mathbf{C}\mathbf{X}_q$. The matrix $\mathbf{C}$ maps aerodynamic strengths to induced tangential velocities. The matrix $\mathbf{M}$ transfers channel flows to the prescribed normal velocities.

These velocities are linear functions of the channel flows plus the impermeable-airfoil contribution. Bernoulli's equation therefore gives quadratic panel pressures. Opening averaging preserves this dependence. The opening pressures and normal kinetic heads can thus be evaluated using precomputed quadratic coefficients. All terms in the discretised pressure relation are retained. The reduction avoids repeated panel solves without introducing a small-flow approximation.

The remaining nonlinear unknowns are the dimensionless channel flows $\mathbf{z}$, defined by

$$\mathbf{q} = \mathbf{S}\mathbf{z}, \qquad \mathbf{S} = V_\infty \,\mathrm{diag}(H_1, \dots, H_M). \tag{21}$$

For each trial $\mathbf{z}$, the reduced aerodynamic model provides the opening pressures and inlet normal kinetic heads. Hydraulic inversion then gives the predicted flow vector $\mathbf{q}_{\mathrm{hyd}}(\mathbf{z})$. The coupled residual is

$$\mathbf{R}(\mathbf{z}) = \mathbf{z} - \mathbf{S}^{-1}\mathbf{q}_{\mathrm{hyd}}(\mathbf{z}) = \mathbf{0}. \tag{22}$$

All channel flows are solved simultaneously because they interact through the external pressure field.

A damped Newton method solves the residual equations [17]. The analytical Jacobian is

$$\mathbf{J}_R = \mathbf{I}_M - \mathbf{S}^{-1}\frac{\partial \mathbf{q}_{\mathrm{hyd}}}{\partial \mathbf{z}}. \tag{23}$$

It is obtained by differentiating the quadratic aerodynamic expressions and implicitly differentiating Eq. (16). The hydraulic derivatives include the dependence of the friction factor on the channel Reynolds number. Derivatives are evaluated within the selected inlet and friction branches.

At iteration n, the Newton correction and flow update are

$$\mathbf{J}_R(\mathbf{z}^{(n)})\Delta\mathbf{z} = -\mathbf{R}(\mathbf{z}^{(n)}), \tag{24}$$

$$\mathbf{z}^{(n+1)} = \mathbf{z}^{(n)} + \eta_n \Delta\mathbf{z}. \tag{25}$$

Armijo backtracking selects $0 < \eta_n \le 1$ using the merit function $\frac{1}{2}\|\mathbf{R}\|_2^2$ [17]. The full nonlinear residual is evaluated at every trial update, including any changes in inlet or friction branch. Ill-conditioned Newton systems are solved in a least-squares sense. A residual-gradient direction is used when needed to obtain descent.

Initial flows are estimated from the impermeable-airfoil pressure field or a nearby converged solution. If Newton iterations fail to satisfy the residual tolerance, a trust-region reflective least-squares method provides a fallback [21]. It

uses the same residual and analytical Jacobian. A least-squares termination flag alone does not establish a coupled solution.

Parameter sweeps may additionally use adaptive angle-of-attack continuation, Reynolds-number continuation, coupling homotopy and pseudo-arclength continuation [22]. These procedures provide starting estimates for the requested operating point. Final acceptance uses a full panel solve and the original hydraulic model. The flow residual and channel energy errors must satisfy the prescribed tolerances. The flow direction and hydraulic regime must also be admissible. Only solutions at the requested operating conditions and full aero–hydraulic coupling are accepted.

2.5. Verification and aerodynamic coefficients

A converged reduced solution is independently evaluated using a full panel solve and the original hydraulic inversion. Acceptance requires a sufficiently small dimensionless flow mismatch, a consistent channel energy balance and an admissible flow direction. Failed cases are recorded as nonconverged.

The pressure-force contribution of panel i is

$$\Delta C_{x,i} = -C_{p,i} n_{x,i} \frac{S_i}{c}. \tag{26}$$

$$\Delta C_{y,i} = -C_{p,i} n_{y,i} \frac{S_i}{c}. \tag{27}$$

The total force coefficients are

$$C_x = \sum_{i=1}^{N} \Delta C_{x,i}. \tag{28}$$

$$C_y = \sum_{i=1}^{N} \Delta C_{y,i}. \tag{29}$$

Lift and drag coefficients follow by rotation through the angle of attack α:

$$C_L = C_y \cos\alpha - C_x \sin\alpha. \tag{30}$$

$$C_D = C_x \cos\alpha + C_y \sin\alpha. \tag{31}$$

Using a clockwise-positive moment convention, the moment coefficient about $(x_{\mathrm{ref}}, y_{\mathrm{ref}})$ is

$$C_M = -\sum_{i=1}^{N} \left[\frac{x_i - x_{\mathrm{ref}}}{c} \Delta C_{y,i} - \frac{y_i - y_{\mathrm{ref}}}{c} \Delta C_{x,i} \right]. \tag{32}$$

These coefficients represent integrated pressure loads over the discretised panel boundary. They exclude viscous shear and an explicit additional pore momentum-flux reaction.

3. Results and discussion

The aerodynamic behaviour of the three porous-airfoil configurations was investigated over an angle-of-attack range of $-5°$ to $15°$. Three pore heights, $H = 8$, 9 and 10 mm, were considered for the horizontal, vertical and horizontal–vertical porous airfoils. The corresponding impermeable (solid) airfoil panel-method solution and the inviscid XFOIL prediction are included as reference cases.

The effect of the pores is first examined through the lift and moment coefficients over the angle-of-attack range. The observed aerodynamic trends are then interpreted using the corresponding pressure and velocity field differences. A detailed flow-field comparison is presented at $\alpha = 10°$ for a pore height of $H = 9$ mm.

3.1. Panel method validation against XFOIL

The impermeable source–vortex panel-method solution is compared with the corresponding inviscid XFOIL prediction in Figures. 2 and 3. Both methods show similar trends in lift and pitching moment over the investigated angle-of-attack range.

This agreement provides a useful reference for assessing the aerodynamic effects introduced by the pores. Small differences between the panel-method and XFOIL predictions are expected because the two approaches use different numerical formulations, although both represent inviscid external flow.

The impermeable panel-method solution is therefore used as the primary reference when evaluating the porous-airfoil configurations. This ensures that the porous and solid airfoils are compared using the same geometry, panel discretisation, aerodynamic formulation and operating conditions.

3.2. Effect of pores on lift

Figure 2 compares the variation of lift coefficient with angle of attack for the three porous-airfoil configurations and pore heights. The horizontal porous airfoil produces the largest change in lift. At positive angles of attack, its lift coefficient is higher than that of the impermeable reference, and the difference becomes more pronounced as the angle of attack increases. The effect also strengthens with pore height, with $H = 10$ mm producing the highest lift at larger positive angles of attack. At negative angles of attack, increasing the pore height similarly increases the magnitude of the negative lift.This nearly symmetric lift response about the zero-lift condition is consistent with the symmetric NACA 0018 geometry.

For the horizontal configuration, both ends of each pore are located on the same airfoil surface. The streamwise variation in surface pressure therefore creates a pressure difference between the two ends and drives internal flow from the locally higher-pressure region towards the lower-pressure region. This produces suction at one end and blowing at the other, changing the local surface-normal velocity and redistributing the external pressure and velocity fields along the airfoil surface. At positive angle of attack, the

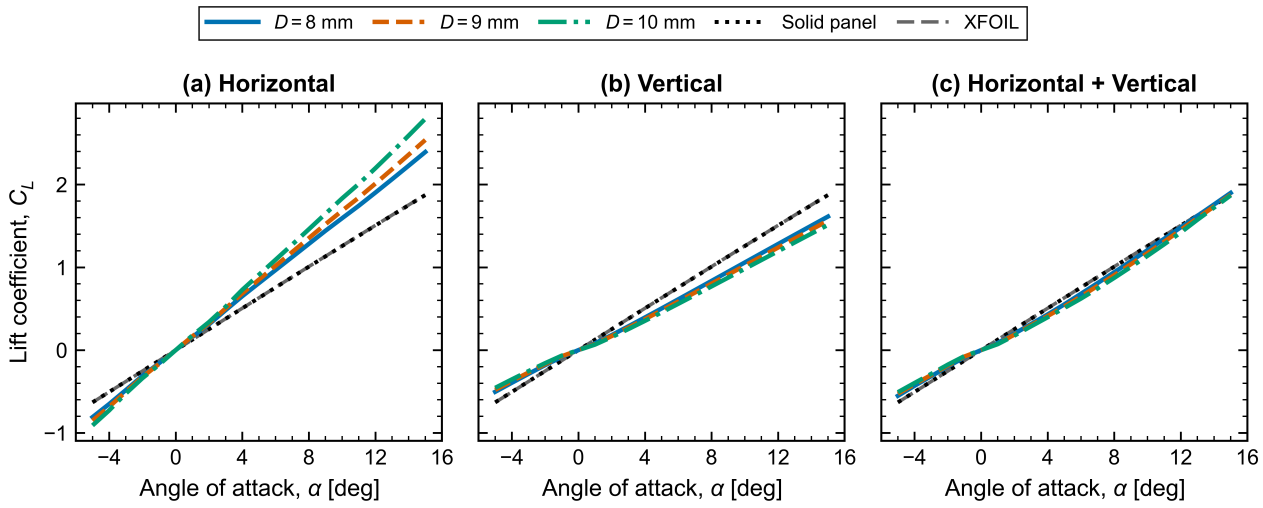


Figure 2: Lift coefficient as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Results are shown for pore heights $H = 8$, 9 and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

different pressure distributions over the suction and pressure surfaces lead to a net redistribution of aerodynamic loading that increases lift.

The stronger dependence on pore height is consistent with the reduction in hydraulic resistance as the pore becomes larger. For a given pressure difference, a larger pore height permits a greater internal flow and therefore a stronger interaction between the pore flow and the external aerodynamic field.

The vertical porous airfoil shows a distinctly different behaviour. At positive angles of attack, its lift coefficient is generally lower than that of the impermeable airfoil, while the three pore-height cases remain relatively close over most of the investigated range.

In this configuration, each pore connects the pressure surface to the suction surface. The resulting pressure difference drives internal flow between the two surfaces and partially reduces the pressure difference responsible for lift. This pressure equalisation is consistent with the lower lift coefficient obtained for the vertical porous airfoil. The small separation between the pore-height cases also indicates that this configuration is less sensitive to pore height than the horizontal arrangement.

The horizontal–vertical porous airfoil remains comparatively close to the impermeable reference. At positive angles of attack, its lift coefficient is generally slightly lower than that of the solid airfoil, and the three pore-height cases remain closely grouped.

This behaviour reflects the simultaneous influence of the horizontal and vertical pore arrangements. The horizontal pores redistribute flow between different chordwise locations on the same surface, whereas the vertical pores transfer flow between the pressure and suction surfaces. The lift-enhancing redistribution associated with the horizontal pores is therefore partly offset by the pressure equalisation introduced by the vertical pores.

The horizontal and vertical pores remain physically separate and hydraulically independent. However, their aerodynamic effects interact through the external pressure field, since changes produced by one set of pores alter the surface pressures that drive the other set. Consequently, the response of the horizontal–vertical porous airfoil is not a simple superposition of the two individual configurations.

Overall, the results show that the aerodynamic effect of the pores is governed mainly by their orientation, placement and the pressure difference between the connected surface locations. Pore height also influences the response, but its effect is strongest for the horizontal configuration and considerably weaker for the vertical and horizontal–vertical arrangements.

3.3. Effect of pores on pitching moment

The corresponding moment coefficients are presented in Figure. 3. The horizontal porous airfoil produces the largest increase from the impermeable-airfoil response. As the positive angle of attack increases, the pitching-moment coefficient becomes progressively more negative. The magnitude of this change also increases with pore height, with $H = 10$ mm producing the largest negative pitching moment at higher positive angles of attack.

This behaviour is associated with the chordwise redistribution of surface pressure caused by the horizontal pores. Since the two ends of each pore are located at different chordwise positions along a given airfoil surface, the pressure changes induced by the internal flow act at different distances from the quarter-chord reference point. These local pressure variations therefore contribute differently to the pitching moment. Increasing the pore height strengthens the internal flow and the resulting pressure redistribution, leading to a larger change in the pitching-moment coefficient.

The increase in lift obtained with the horizontal porous airfoil is therefore accompanied by a larger negative pitching

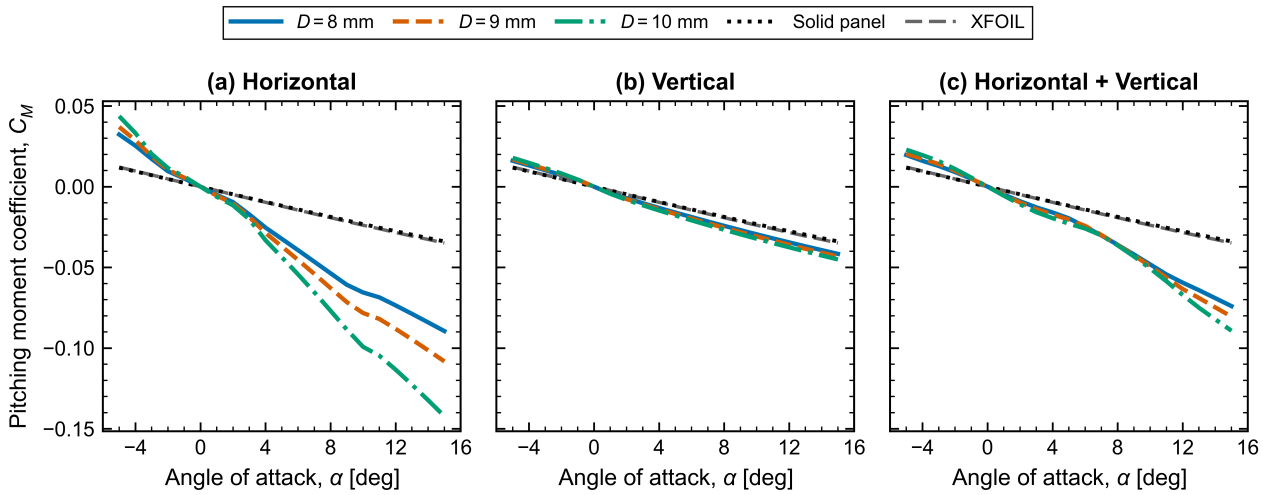


Figure 3: Pitching-moment coefficient about the quarter chord as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Results are shown for pore heights $H = 8$, 9 and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

moment. This represents an important aerodynamic trade-off, since the configuration that provides the greatest lift enhancement also produces the largest change in longitudinal loading.

The vertical porous airfoil produces considerably smaller changes in moment, and the three pore-height cases remain closely grouped over most of the investigated angle-of-attack range. This behaviour is consistent with the weaker dependence on pore height observed in the lift results.

For the vertical configuration, the two ends of each pore are positioned on the upper and lower surfaces at approximately the same chordwise location. The resulting pressure redistribution therefore occurs mainly between the pressure and suction surfaces rather than between widely separated positions along the chord. The corresponding change in the chordwise pressure distribution is smaller than for the horizontal pores, resulting in a weaker pitching-moment response.

The horizontal–vertical porous airfoil exhibits behaviour between that of the individual horizontal and vertical configurations. The pitching moment becomes increasingly negative with positive angle of attack, although the departure from the impermeable-airfoil solution remains smaller than for the horizontal porous airfoil. The influence of pore height also becomes more noticeable at higher positive angles of attack.

In this configuration, the horizontal pores redistribute loading between different chordwise positions, while the vertical pores mainly redistribute pressure between the upper and lower surfaces. The two pore arrangements remain physically separate and hydraulically independent, but their aerodynamic effects interact through the external pressure field. The resulting pitching-moment variation therefore reflects the combined influence of these two pressure-redistribution mechanisms.

Overall, the pitching-moment results show that the effect of the pores depends not only on the extent of the pressure redistribution, but also on where it occurs along the chord. The horizontal porous airfoil produces the largest change in moment because the associated pressure variations occur at widely separated chordwise locations, creating a stronger moment about the quarter-chord reference point. In contrast, the vertical porous airfoil produces a smaller change because its pressure redistribution occurs mainly between the upper and lower surfaces at similar chordwise positions.

3.4. Flow field characteristics of porous airfoils

To examine how the pores alter the external flow, the pressure and velocity fields of the porous airfoils are compared with those of the corresponding impermeable airfoil at $\alpha = 10°$ and $H = 9$ mm.

The comparison is presented using pressure- and velocity-difference contours rather than the absolute flow fields. The absolute contours are dominated by the overall aerodynamic behaviour of the airfoil and therefore make the local influence of the pores more difficult to identify. Subtracting the impermeable-airfoil solution isolates the changes introduced by the pore flow, allowing the location, direction and relative strength of the resulting pressure and velocity variations to be examined more clearly.

3.4.1. Pressure field characteristics

Figure 4 presents the pressure difference between the porous and impermeable airfoils,

$$\Delta p = p_{\text{porous}} - p_{\text{solid}}, \tag{33}$$

for the three pore arrangements.

The pressure difference is shown instead of the absolute pressure field to isolate the changes introduced by the pores. In the absolute solution, the pressure distribution

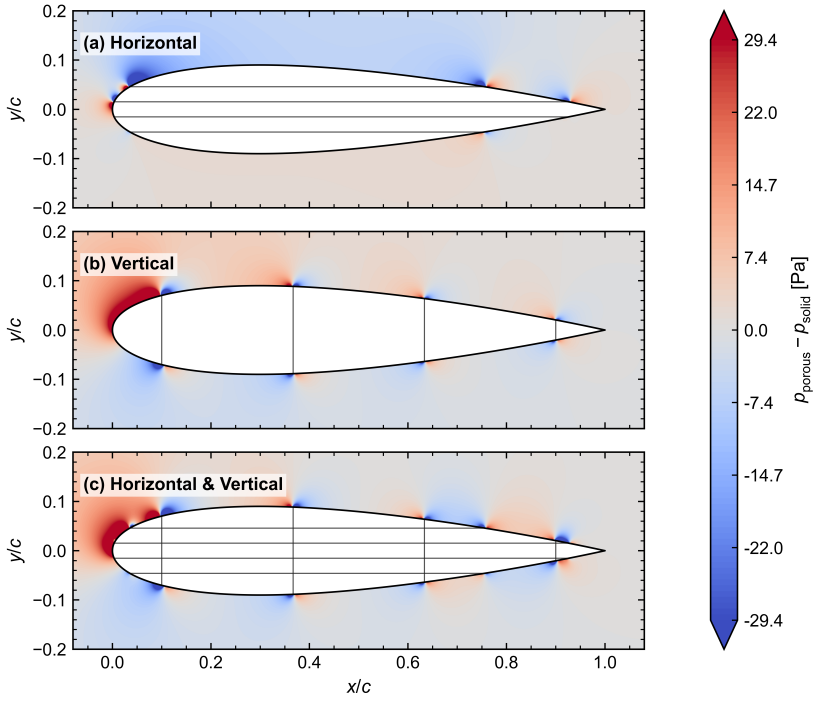


Figure 4: Pressure-field difference between the porous and impermeable airfoils at $\alpha = 10°$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in pressure relative to the impermeable airfoil, while negative values indicate a reduction.

is dominated by the overall airfoil loading and the strong pressure gradients near the leading edge. Subtracting the impermeable-airfoil solution makes the local changes associated with the pore flow easier to identify.

The strongest pressure differences occur near the pore ends, where the internal flow directly influences the surrounding external flow. In several locations, neighbouring positive and negative values of Δp form a dipole-like pattern. This is associated with suction at one end of a pore and blowing at the other, producing local regions of reduced and increased pressure relative to the impermeable airfoil.

The effect of these pressure changes on lift depends on where they occur on the airfoil. At positive angle of attack, a reduction in pressure on the upper suction surface strengthens the suction and contributes to higher lift, whereas an increase in pressure has the opposite effect. On the lower pressure surface, an increase in pressure contributes positively to lift, while a pressure reduction tends to decrease it.

For the horizontal porous airfoil, the pressure differences are mainly concentrated around the chordwise pore ends. Because both ends of each pore lie on the same airfoil surface, the internal flow redistributes pressure between different chordwise locations. The combined pressure changes on the upper and lower surfaces increase the pressure difference across the airfoil, consistent with the higher lift coefficient observed for this configuration.

The vertical porous airfoil produces a different pattern because each pore connects the pressure surface to the suction surface. Flow is therefore driven from the higher-pressure region towards the lower-pressure region, producing local pressure changes at both pore ends. At the same time, this cross-surface flow partially reduces the pressure difference between the two surfaces, which is consistent with the lower lift coefficient obtained for the vertical configuration.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores redistribute pressure along the airfoil surfaces, while the vertical pores transfer flow between the pressure and suction surfaces. Although the two pore systems remain physically separate and hydraulically independent, their aerodynamic effects interact through the external pressure field. Consequently, the pressure equalisation associated with the vertical pores partly counteracts the pressure redistribution produced by the horizontal pores, giving an overall aerodynamic behaviour closer to that of the impermeable airfoil.

3.4.2. Velocity field characteristics

The velocity field is presented as the percentage difference relative to the impermeable-airfoil solution,

$$\Delta|\boldsymbol{V}|_{\%} = \frac{|\boldsymbol{V}|_{\text{porous}} - |\boldsymbol{V}|_{\text{solid}}}{|\boldsymbol{V}|_{\text{solid}}} \times 100. \tag{34}$$

This comparison isolates the changes introduced by the pores from the dominant background flow around the airfoil. The largest velocity differences occur close to the pore ends, where the imposed surface-normal flow locally perturbs the

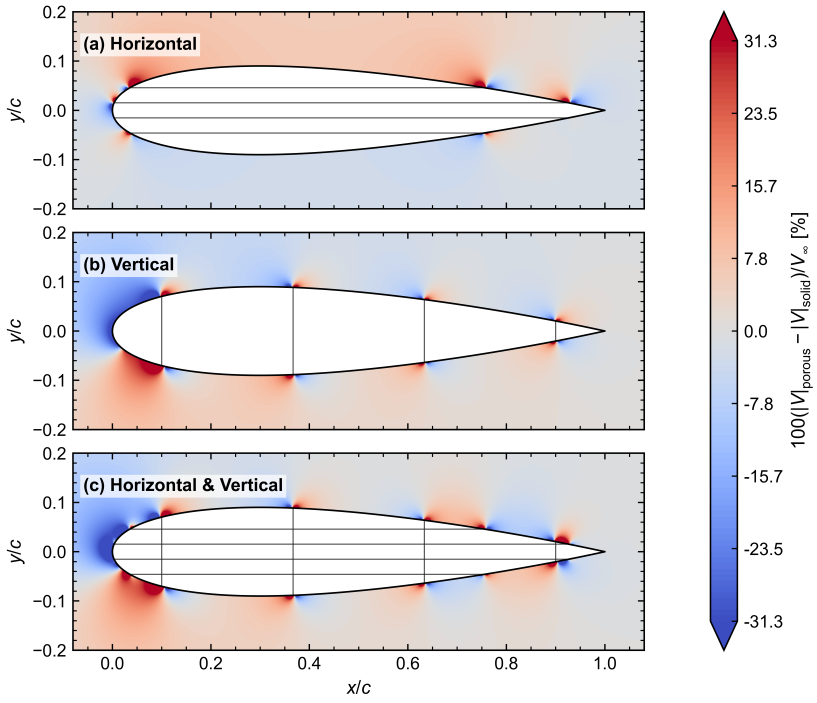


Figure 5: Percentage difference in velocity magnitude between the porous and impermeable airfoils at $\alpha = 10°$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in velocity magnitude, while negative values indicate a reduction relative to the impermeable airfoil.

external velocity field. Adjacent regions of acceleration and deceleration are visible around several pores and correspond closely to the pressure patterns shown in Figure. 4.

The velocity and pressure fields show the expected inverse relationship described by Bernoulli's equation. Regions where the flow accelerates are generally associated with a reduction in static pressure, whereas regions of lower velocity correspond to higher static pressure. The velocity contours therefore help explain how the pore-induced changes in the local flow field lead to the observed pressure redistribution and, consequently, changes in the aerodynamic forces and pitching moment.

For the horizontal porous airfoil, alternating regions of acceleration and deceleration are concentrated around the chordwise pore ends. Since both ends of each pore lie on the same airfoil surface, the internal flow redistributes the external velocity between different chordwise locations. The combined changes on the suction and pressure surfaces strengthen the pressure difference across the airfoil, consistent with the increase in lift and the larger change in pitching moment observed for this configuration.

The vertical porous airfoil shows a different velocity pattern. Flow between the pressure and suction surfaces changes the external velocity on both sides of the airfoil and reduces part of the velocity contrast between them. This behaviour is consistent with the partial pressure equalisation discussed previously and the corresponding reduction in lift. A broader region of velocity difference is also visible near the leading edge, where the external flow experiences strong pressure and velocity gradients.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores produce local chordwise acceleration and deceleration, while the vertical pores redistribute flow between the pressure and suction surfaces. Although the two pore arrangements remain physically separate and hydraulically independent, their aerodynamic effects interact through the external flow field. The resulting velocity distribution therefore reflects the combined influence of both mechanisms rather than a simple superposition of the individual configurations.

Overall, the velocity contours support the trends observed in the pressure field and aerodynamic coefficients. The horizontal pores primarily redistribute the flow along the airfoil surfaces, whereas the vertical pores transfer flow between the pressure and suction surfaces. These different flow mechanisms contribute to the contrasting lift and pitching-moment behaviour of the three porous-airfoil configurations.

4. Conclusion and future work

A reduced-order aero–hydraulic framework has been developed to investigate NACA 0018 airfoils containing pressure-driven internal channels. The external flow is represented using a source–vortex panel method and is coupled to finite-length rectangular-channel models through the pore-normal velocity. The main conclusions of the present study are:

- The aerodynamic response of a porous airfoil depends strongly on the orientation, location and connectivity of the internal channels. Porosity alone is therefore insufficient to describe the resulting aerodynamic behaviour.
- Horizontal chordwise channels produce the strongest aerodynamic modification among the configurations considered. By connecting different chordwise locations on the same airfoil surface, these channels redistribute pressure without directly equalising the pressure and suction surfaces.
- At positive angles of attack, the horizontal configuration increases lift relative to the impermeable airfoil. The magnitude of this effect increases with channel height, demonstrating the sensitivity of the aerodynamic response to the hydraulic resistance of the internal passages.
- The lift enhancement produced by the horizontal channels is accompanied by a more negative pitching moment. This demonstrates that improvements in lift cannot be considered independently of changes in longitudinal aerodynamic loading.
- Vertical channels exhibit a fundamentally different behaviour. Connecting the pressure and suction surfaces allows pressure-driven flow between the two sides of the airfoil, producing partial pressure equalisation and consequently reducing lift relative to the impermeable configuration at positive angles of attack.
- The combined horizontal–vertical configuration does not behave as a simple superposition of the two individual arrangements. Although the channel systems are hydraulically independent, they interact through the external pressure field. The pressure equalisation associated with the vertical channels therefore partly offsets the lift-enhancing effect of the horizontal channels.
- Pressure- and velocity-difference fields show that the strongest aerodynamic modifications occur locally around the channel openings. Suction and blowing generated by the internal flow alter the surrounding external velocity and pressure fields, which subsequently modify the integrated aerodynamic forces and pitching moment.
- The results demonstrate that the pressure difference between connected surface locations is a key driver of porous-airfoil behaviour. Channel placement, orientation and height should therefore be considered simultaneously when designing internal porous networks.
- The developed framework provides a computationally inexpensive method for screening multiple internal-channel configurations before progressing to substantially more expensive viscous CFD simulations or experimental investigations.

The present results should be interpreted as inviscid aerodynamic trends. The external panel method does not resolve boundary layers, transition, skin-friction drag, flow separation or stall, and the pressure-based streamwise-force coefficient should therefore not be interpreted as the total drag of the porous airfoil. These limitations define the next stage of the investigation.

4.1. Future work

Future development will focus on:

- incorporating a viscous boundary-layer model into the panel-method framework to estimate skin-friction drag, transition and the influence of viscous–inviscid interaction;
- extending the design space beyond the three prescribed configurations by optimising channel location, orientation, height and connectivity to identify improved internal porous-network topologies;
- investigating selected configurations using higher-fidelity unsteady CFD to resolve pore-mouth flow, boundary-layer interaction, separation and wake dynamics that cannot be represented by the present reduced-order formulation;
- validating the predicted pressure redistribution and aerodynamic trends experimentally using manufactured porous-airfoil models; and
- extending the analysis towards three-dimensional porous wings and application-relevant operating conditions once the underlying aero–hydraulic mechanisms have been established.

The present framework therefore provides the first stage of a hierarchical design process in which rapid reduced-order screening is used to identify promising internal-channel arrangements for subsequent optimisation, high-fidelity viscous simulation and experimental validation.

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